

Longitudinal Biomarker Analysis

- **Urea Nitrogen (Top-left)**
 - **Blue (Longitudinal Phenotype 1):**
 - Starts at moderate levels (~20–25 units).
 - Shows a steady decline over time, reaching near 0 by the end of the trajectory.
 - Variation decreases initially, then broadens slightly as levels approach zero.
 - **Orange (Longitudinal Phenotype 2):**
 - Starts slightly higher (~25–30 units).
 - Shows a gradual increase over time, ending above the starting point (~30–35 units).
 - Variation is relatively wide initially but stabilizes as time progresses.
 - **Interpretation:**
 - Phenotype 1 may represent patients whose kidney function is deteriorating or being treated effectively, leading to decreasing urea.
 - Phenotype 2 indicates a subgroup with increasing urea levels, possibly reflecting progressive metabolic or renal stress.
- **Hematocrit (Top-right)**
 - **Blue (Phenotype 1):**
 - Starts around 40%.
 - Shows a continuous decline over time, reaching below 20%.
 - Variation remains high initially, reflecting heterogeneity, then narrows as most patients drop to lower values.
 - **Orange (Phenotype 2):**
 - Remains relatively constant (~35–40%) across the time course.
 - Very little variation, indicating stable hematocrit levels in this phenotype.
 - **Interpretation:**
 - Phenotype 1 may indicate patients developing anemia or blood loss over time.
 - Phenotype 2 patients maintain normal hematocrit, representing a more stable subgroup.
- **Potassium (Bottom-left)**
 - **Blue (Phenotype 1):**
 - Starts ~4 units.
 - Declines steadily, reaching below 0 by the end.
 - Variation is moderate early on, then tightens as the trend progresses downward.
 - **Orange (Phenotype 2):**
 - Nearly flat (~4 units) with minimal variation.
 - Potassium remains stable throughout.
 - **Interpretation:**

- Phenotype 1 shows progressive hypokalemia, potentially associated with treatment, diet, or kidney issues.
 - Phenotype 2 represents stable potassium levels over time.
- **Sodium (Bottom-right)**
 - **Blue (Phenotype 1):**
 - Starts ~135–140 units.
 - Declines continuously, falling below normal range (~120 units), indicating hyponatremia development.
 - Variation broadens slightly as values drop.
 - **Orange (Phenotype 2):**
 - Essentially flat (~140–145 units) with negligible variation.
 - Sodium levels remain in the normal range.
 - **Interpretation:**
 - Phenotype 1 indicates patients with progressive electrolyte imbalance.
 - Phenotype 2 maintains homeostasis.
- **Overall Analysis**
 - **Phenotype 1 (Blue):**
 - Characterized by progressive decline across all measured biomarkers (Urea, Hematocrit, Potassium, Sodium).
 - Suggests a subgroup of patients with deteriorating metabolic and hematologic status, potentially at higher clinical risk.
 - High clinical relevance for monitoring interventions.
 - **Phenotype 2 (Orange):**
 - Biomarkers are stable or slightly increasing (Urea rises slightly, others flat).
 - Represents a more stable subgroup with normal or mildly changing laboratory profiles.
 - Could serve as a reference/control phenotype.
- **Key Insight:**
 - The two longitudinal phenotypes clearly separate patients by risk trajectories: declining biomarker trends vs. stable trends, which can guide personalized monitoring or interventions.
 - Urea shows a distinct divergence (increase in orange vs. decrease in blue), highlighting it as a potential early indicator for phenotype differentiation.